Chapter 1 – Fourier Transforms

MT3506 – Techniques of Applied Mathematics

Lecture 1:

- 1.1 Definitions

Lecture 2:

- 1.2 Fundamental relations
- 1.3 FT of delta & Heaviside functions

Lecture 3:

- 1.4 Computation of Fourier Transforms

Lecture 4:

- 1.4 Computation of Fourier Transforms (ctd)
- **1.5 Convolutions**

Next time:

- 1.5 Convolutions (ctd)
- 1.6 Energy Theorems

$$\underbrace{\tilde{f}(\omega)}_{\text{frequency}} = \int_{-\infty}^{\infty} \underbrace{f(t)}_{\text{time}}\, e^{-i\omega t} dt$$

Image processing: Digital filters

zoom

1.5 Convolutions

Definition. Let f and g be two functions defined on the real line. We define their convolution $f * g$ as

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t')g(t - t')dt'$$

Example:

1.5 Convolutions

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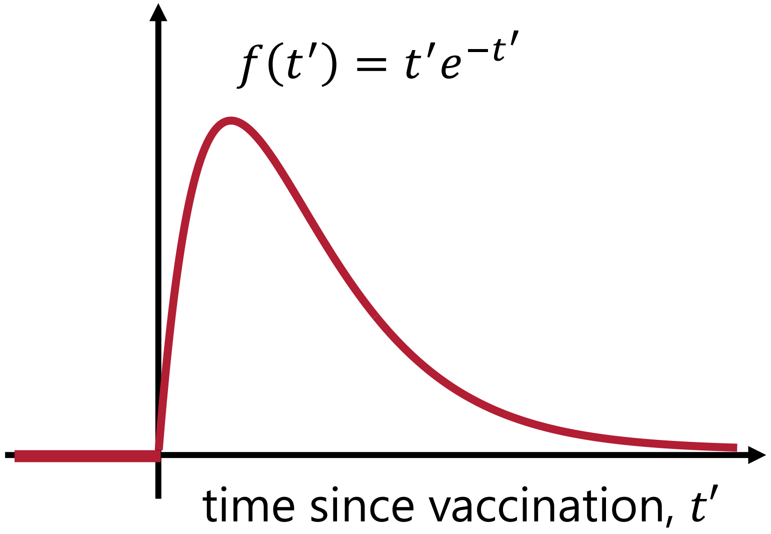
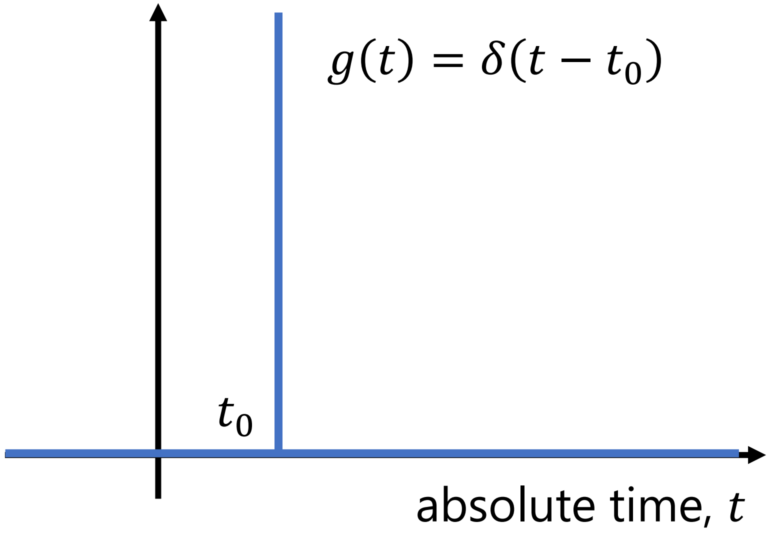
Example:

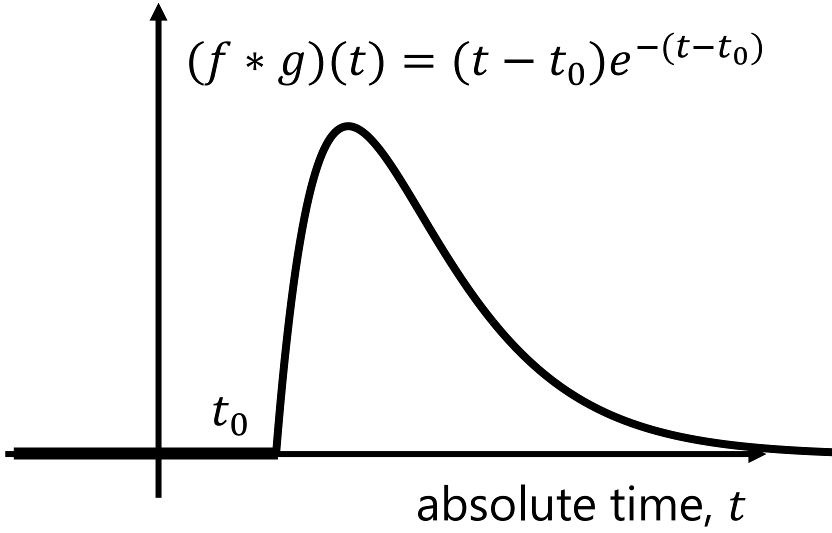
Post-vaccine response, $f(t')$



Vaccination history, $g(t)$



Antibody level, $(f * g)(t)$



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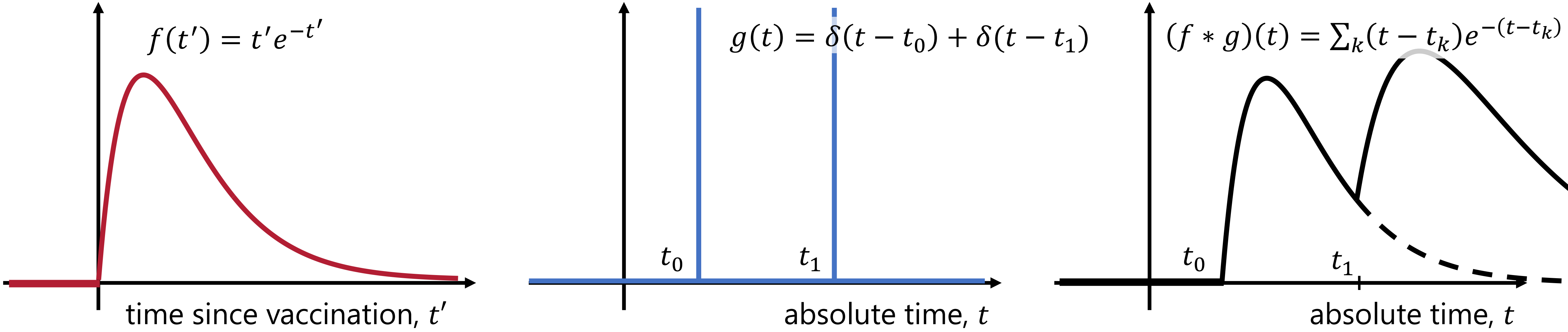

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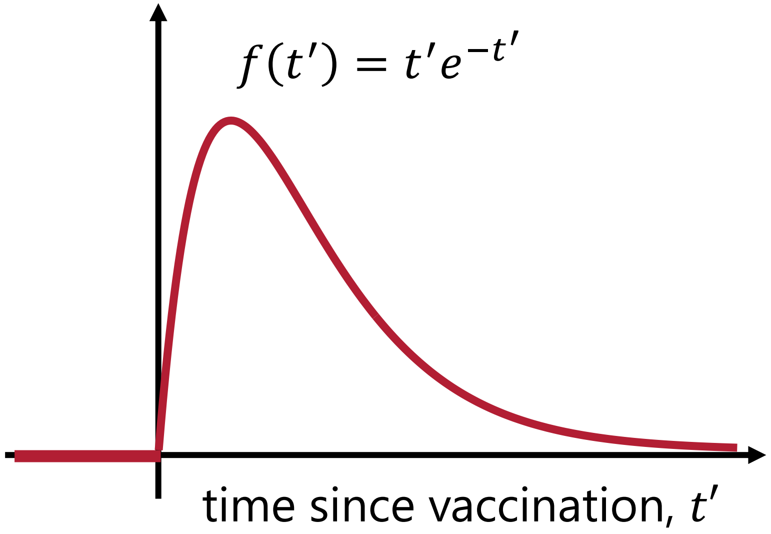
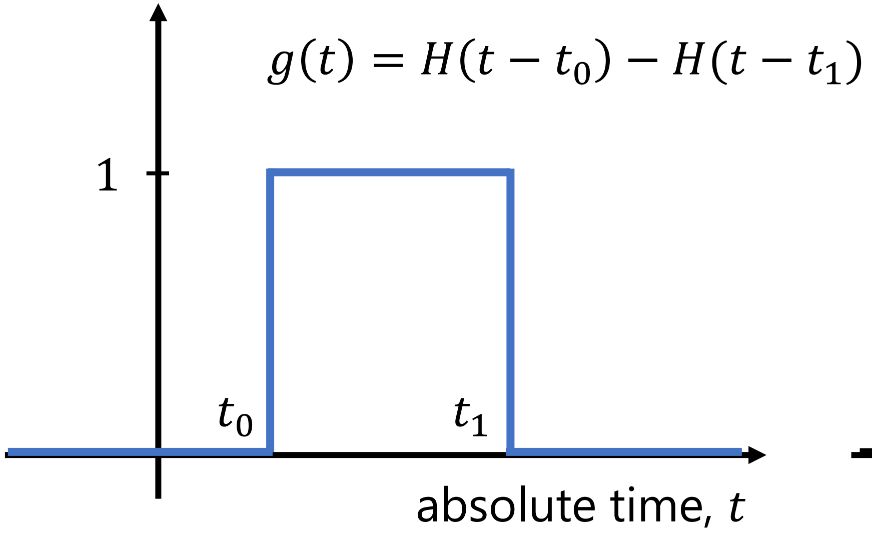
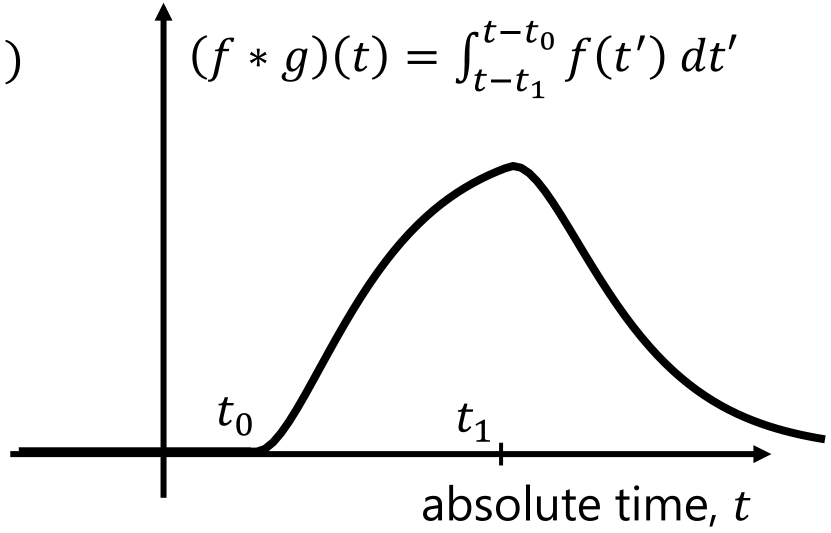


Vaccination programme, $g(t)$



e.g., seasonal vaccination

Antibody level, $(f * g)(t)$



*population-averaged

1.5 Convolutions

Theorem (the convolution theorem). Suppose $\tilde{f}$, $\tilde{g}$, and $f * g$ exist (i.e., the integrals are all well-defined). Then $\widetilde{f * g}$ also exists and

$$\widetilde{f * g} = \tilde{f}\tilde{g}$$

Proof:

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[exchanging order of integration] $$= \int_{-\infty}^{\infty} f(t') \int_{-\infty}^{\infty} g(t - t')e^{-i\omega t} dt\, dt'$$

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[rearranging integrand]

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[substituting $u = t - t'$] $= \int_{-\infty}^{\infty} f(t') e^{-i\omega t'} \boxed{\int_{-\infty}^{\infty} g(u) e^{-i\omega u} du}\, dt' \quad = \tilde{f}(\omega)\tilde{g}(\omega)$ □

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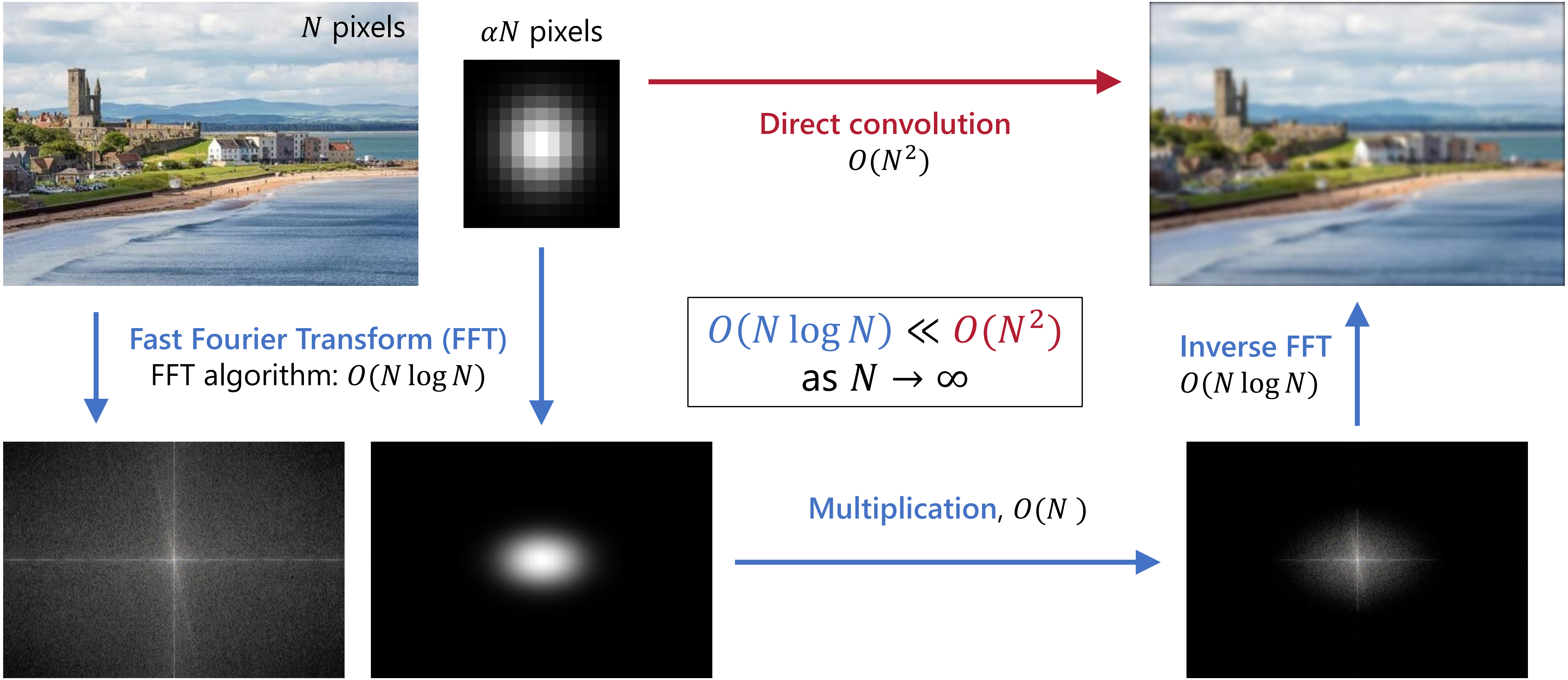
The convolution theorem is the **backbone of fast convolution algorithms**.

Applications: audio processing, image processing, large-integer multiplication.



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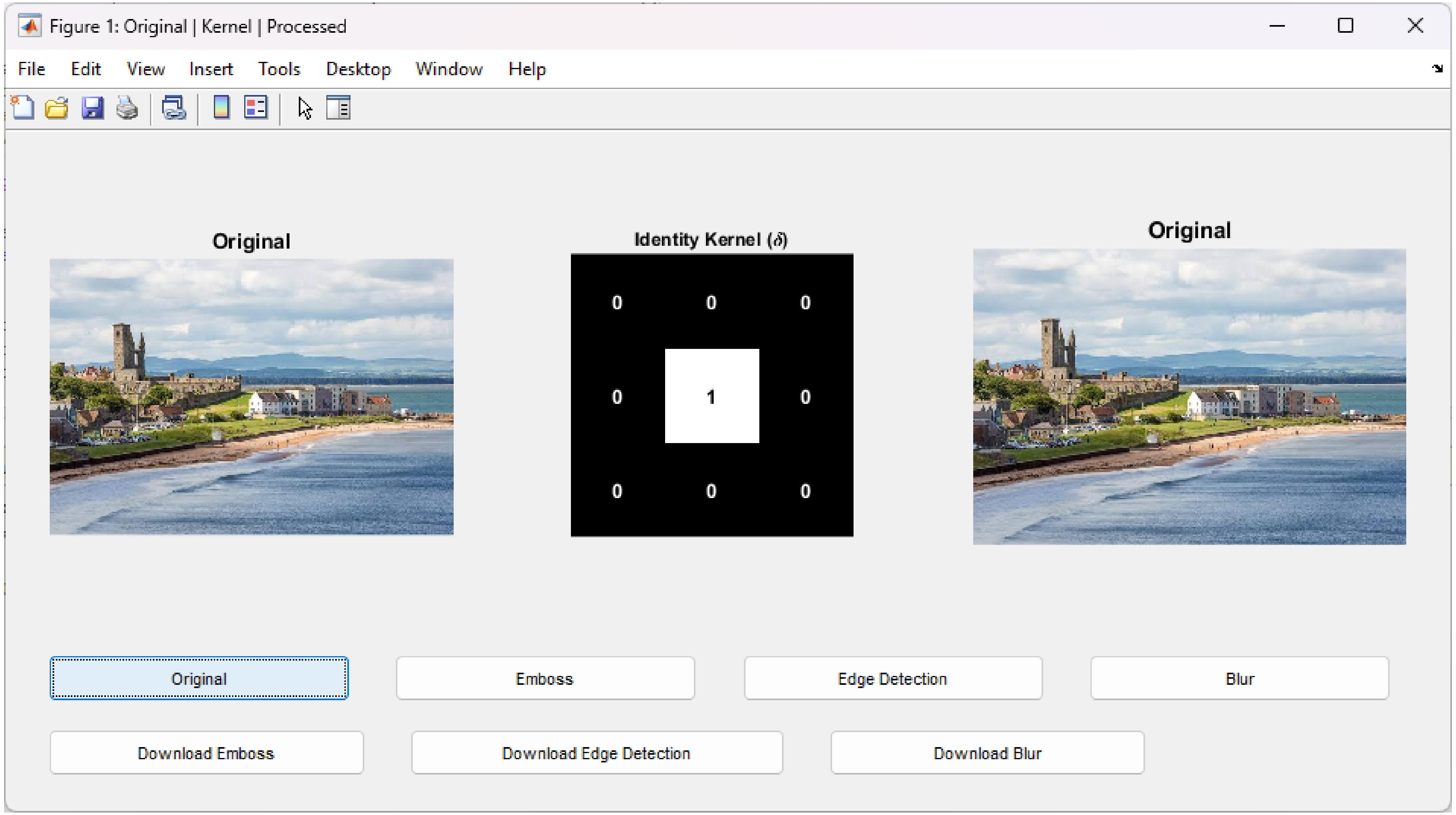
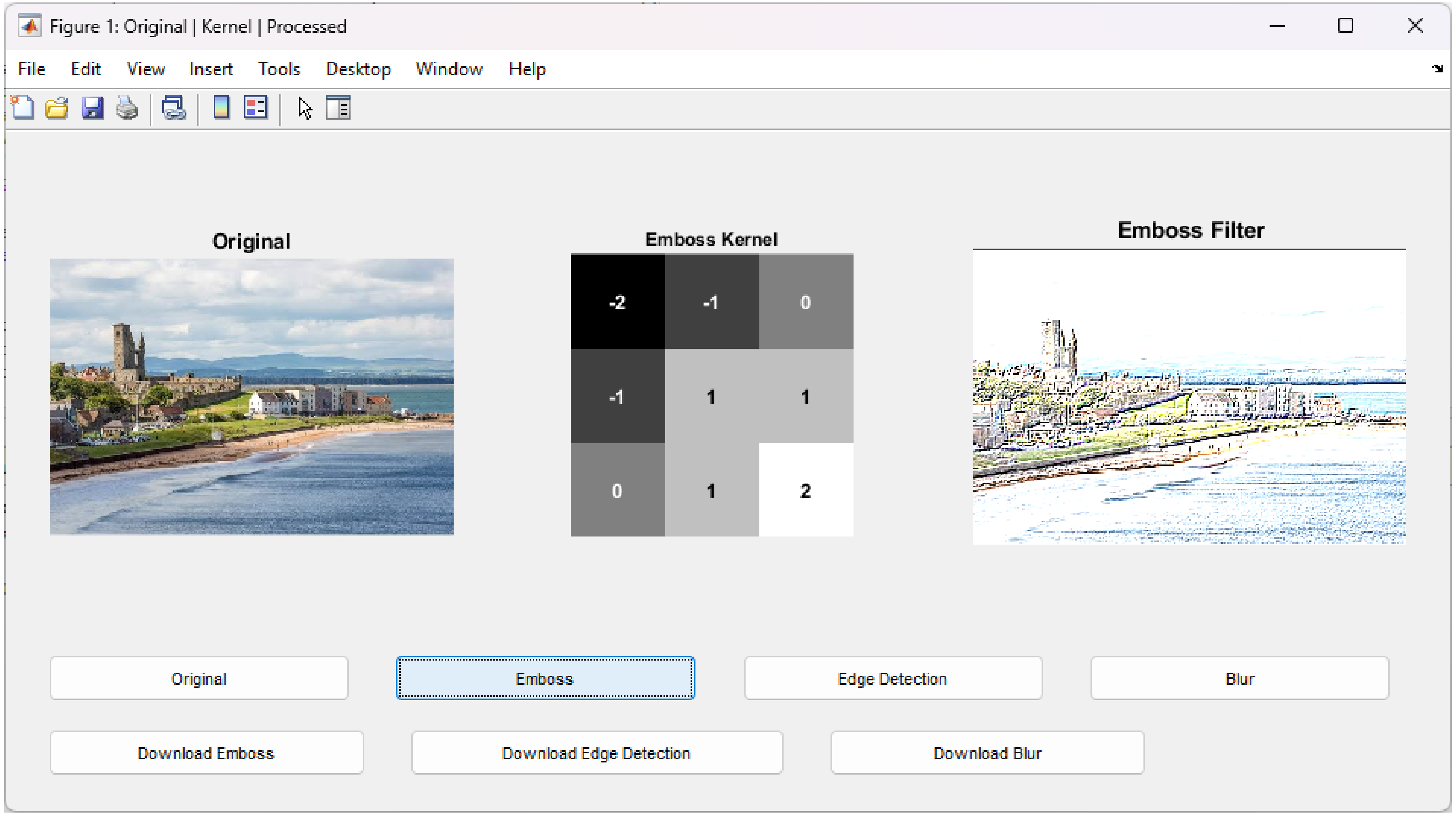


FFT demo (MATLAB)

1.5 Convolutions



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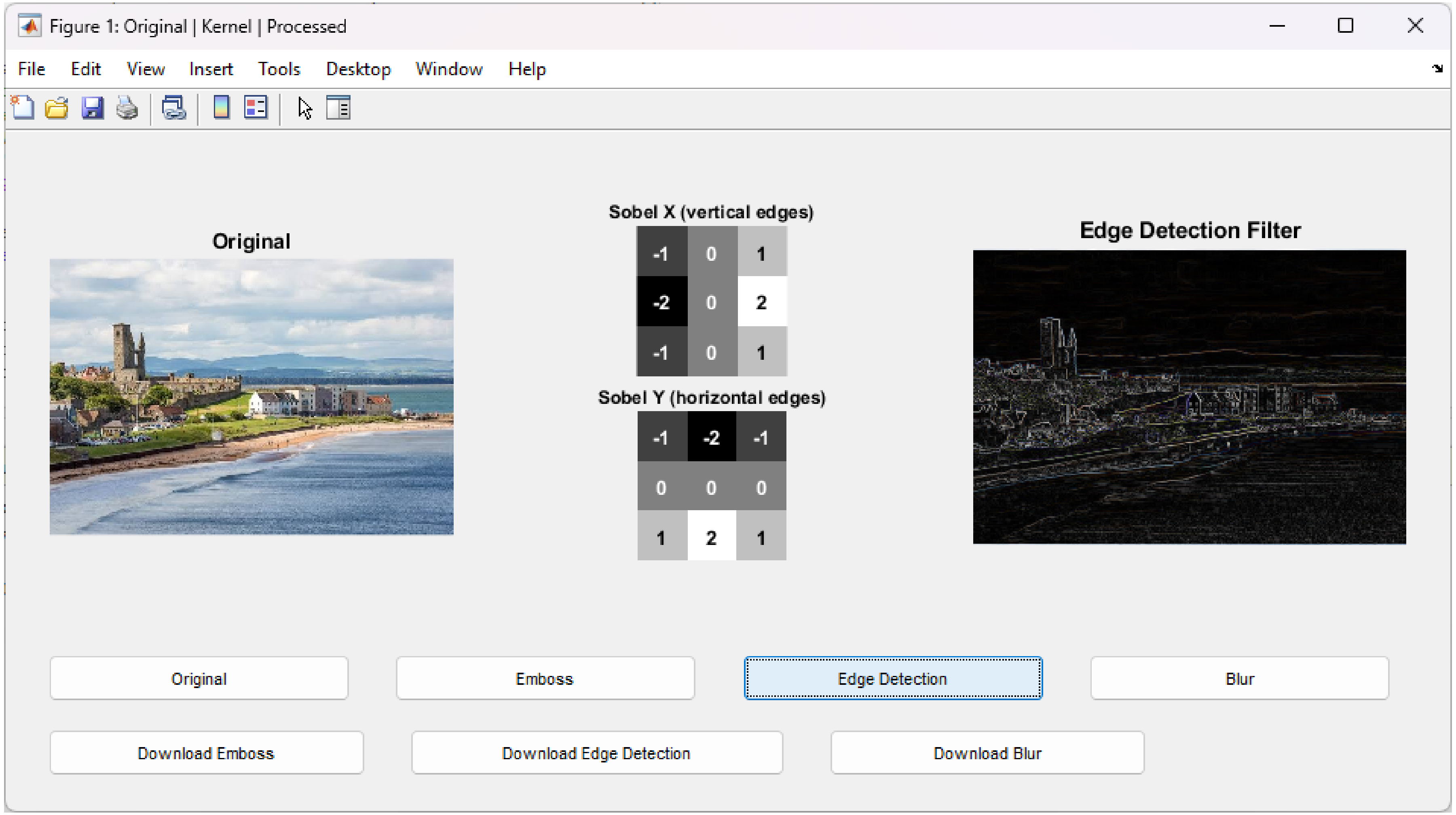

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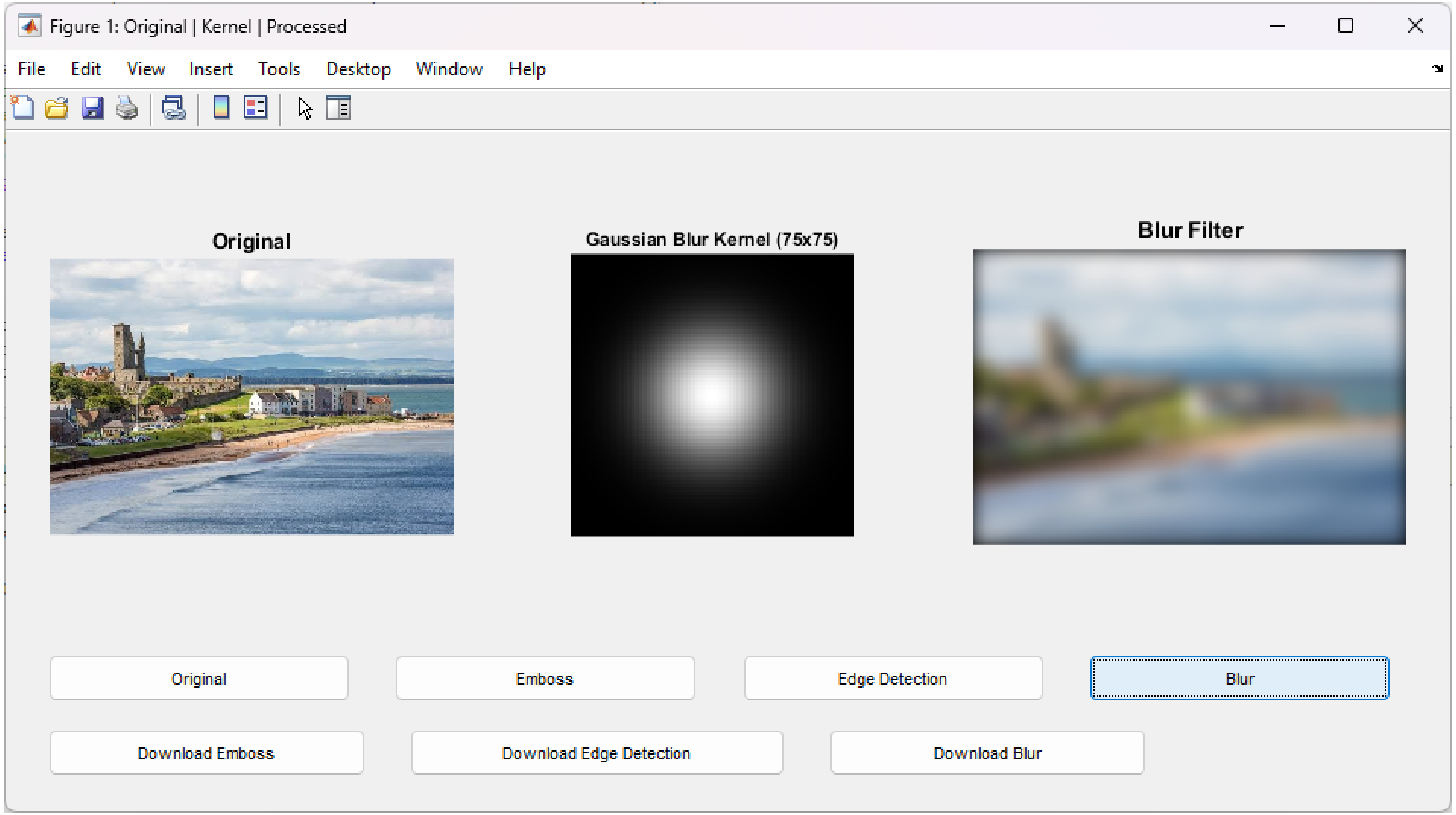

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